

RUI QIN

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🎓 EDUCATION

Zhejiang University, Hangzhou, China 2023 – Present

PhD student in Pharmacy, expected June 2028

M.S. student in AI Pharmacy (2023 – 2025); transferred to PhD track

Sun Yat-Sen University, Guangzhou, China 2019 – 2023

B.S. in Pharmaceutical Sciences

🔬 RESEARCH EXPERIENCE

Large Language Model Systems for Drug Design 2023 – Present

PhD Thesis Project Zhejiang University

- Developed chemical language model frameworks based on autoregressive and diffusion-based architectures as foundational model for multi-task drug design scenarios.
- Applied chemical language models to task-specific drug discovery problems, including molecule generation, optimization, and property prediction.

Application of Molecular Generative Models in Drug Design 2021 – Present

Research Project Zhejiang University & Sun Yat-Sen University

- Applied molecular generative models to practical drug design scenarios and evaluated generated molecules.
- Developed benchmarks for molecular generation using virtual screening, cheminformatics evaluation, and drug-likeness analysis.

📄 SELECTED PUBLICATIONS

1. **Rui Qin***, Zijie Chen*, Yurong Li, et al. *Revisiting Target-Aware de novo Molecular Generation with TarPass: Between Rational Design and Texas Sharpshooter*. **Advanced Science**, 2026.
2. Jike Wang*, **Rui Qin***, Mingyang Wang*, et al. *Token-Mol 1.0: tokenized drug design with large language models*. **Nature Communications**, 2025.
3. Jike Wang*, Hao Luo*, **Rui Qin***, et al. *3DSMILES-GPT: 3D molecular pocket-based generation with token-only large language model*. **Chemical Science**, 2025.
4. **Rui Qin**, Hao Zhang, Weifeng Huang, et al. *Deep learning-based design and screening of benzimidazole-pyrazine derivatives as adenosine A_{2B} receptor antagonists*. **Journal of Biomolecular Structure and Dynamics**, 2023.

🏆 SELECTED HONORS AND FUNDING

Funding:

Scientific Research Fund of Zhejiang Provincial Education Department, Principal Investigator 2025

Honors and Awards:

Honor for Graduate-Excellence in Academic Innovation, Zhejiang University 2025

Wong Lo Kat Scholarship, School of Pharmaceutical Sciences, Sun Yat-Sen University 2023

Second-Class Academic Scholarship, Sun Yat-Sen University 2022

Gold Award, China International “Internet+” College Students Innovation and Entrepreneurship Competition, Guangdong Provincial Competition, Team Member 2022

🗨 PRESENTATIONS

Oral Presentation, ACS Spring 2026

Mar. 2026

Virtual Graduate Students Symposium in Asia-Pacific Region on Computational Chemistry; *Excellent Presentation Award*

Poster Presentation, Chinese Chemical Society Conference on Computational Chemistry May 2025

⚙ SKILLS

- Programming: Python, PyTorch, Linux
- Machine Learning: Large Language Models, Representation Learning
- Drug Design: Molecular Generation, Virtual Screening, Docking (Schrödinger, Vina), Cheminformatics, Molecular Dynamic (Gromacs, Amber)
- Toolkits: RDKit, Huggingface Transformers, PyTorch Geometric, OpenBabel, PyMOL